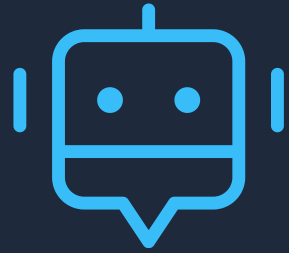


AI Chatbot & RAG PDF Chatbot

Business-Oriented AI Solutions



PROJECT PORTFOLIO



AI Chatbot

Developed a conversational AI assistant capable of generating human-like responses and maintaining context-aware interactions.



RAG PDF Chatbot

Built an intelligent document assistant that retrieves information from PDF files and provides accurate AI-generated answers.



Project Goal

Improve communication efficiency and information retrieval through practical AI-powered solutions.

AI CHATBOT

Project overview

Developed an AI-powered chatbot capable of generating human-like responses and maintaining context-aware conversations. The system supports multi-turn interactions, enabling more natural and engaging user experiences.

Technology Stack



BUISSNESS PURPOSE



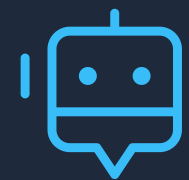
Faster responses

Provides instant responses to user queries, reducing waiting time.



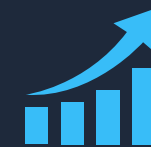
Better Customer Support

Improves customer interaction through 24/7 availability.



Process Automation

Reduces repetitive communication tasks through automation.



Improved Efficiency

Enhances productivity by handling multiple conversations simultaneously.

KEY FEATURES

 **AI-Generated Responses**

 **Multi-Turn Conversations**

 **Chat History
Management**

 **Adjustable AI Settings**

My Contributions



Developed Chatbot Logic



Integrated Groq API



**Managed Conversation
Flow**



Tested AI Parameters



**Improved Response
Quality**

Challenges & Solutions

Challenges

 API Integration

 Chatbot Memory

 Response Quality

Solutions

Studied API documentation and tested different request formats.

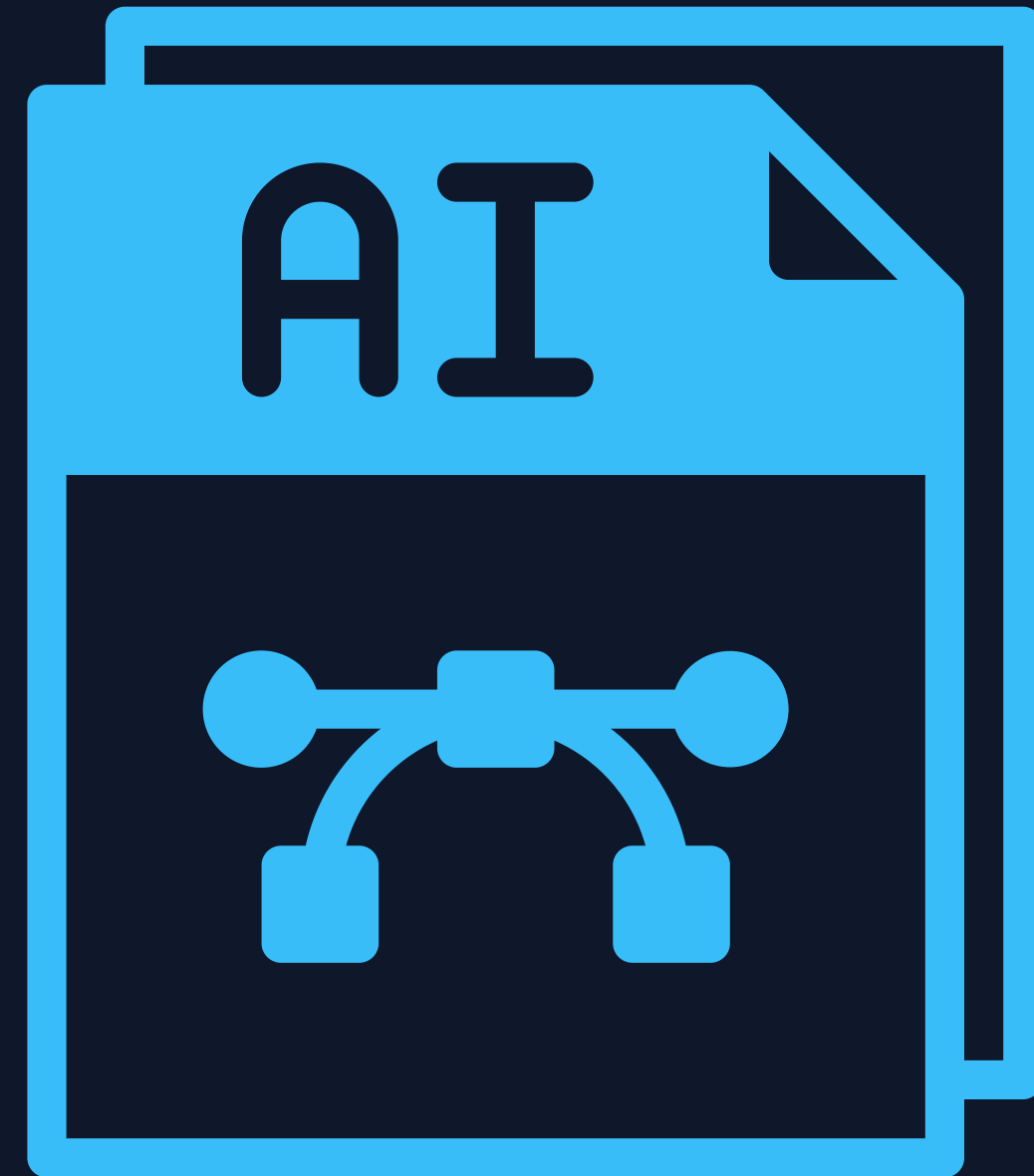
Implemented structured chat history management.

Experimented with temperature and max_tokens settings.

RAG PDF CHATBOT

Project overview

Developed a RAG-based chatbot capable of answering questions from PDF documents. The system retrieves relevant information from uploaded files and generates accurate AI-powered responses.



BUISNESS PURPOSE



Faster Information Retrieval

Users can quickly find relevant information from large PDF documents.



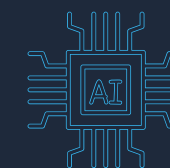
Reduced Manual Searching

Eliminates the need to manually browse through lengthy documents.



Improved Productivity

Saves time by providing instant answers from uploaded PDFs.



Intelligent Document Interaction

Allows users to interact with documents using natural language questions.

KEY FEATURES

 **PDF Upload Support**

 **Text Extraction**

 **Text Chunking and
Embedding generation**

 **ChromaDB Storage**

My Contributions

- Implemented PDF Processing
- Built Retrieval Pipeline
- Integrated ChromaDB
- Connected LLM
- Designed Streamlit Interface

Challenges & Solutions

Challenges



ChromaDB Permission Issues



Streamlit Rerun Handling



Managing Uploaded PDFs

Solutions

Improved project structure and database configuration.

Implemented better session and state management.

Optimized PDF processing and file handling workflow..